

Enhanced UAV Classification: Gaining Deeper Insights from Mechanical Vibration Over RF Characteristics

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Abstract—UAV classification plays a critical role in enhancing authentication and authorization within UAV-based aerial networks, providing a robust mechanism to prevent unauthorized access and ensure secure operations. This paper presents a novel approach to UAV classification by combining RF characteristics and mechanical vibration data, exploring how these methods can strengthen the security and reliability of UAV communication systems. Vibration data, captured through onboard accelerometers, is investigated as a complement to traditional RF fingerprinting, offering an alternative for UAV identification in dynamic environments where RF-based methods may face limitations. Deep learning models, including a Recurrent Neural Network (RNN) for vibrational data and a Convolutional Neural Network (CNN) for 5G communication data, were developed to classify UAVs with high accuracy. The results demonstrate that integrating mechanical vibration data significantly enhances classification performance compared to using RF fingerprinting alone, providing new avenues for secure UAV authentication and authorization. Furthermore, this study emphasizes the importance of efficient, real-time classification methods for deployment in 5G networks, ensuring both the security and operational integrity of UAV systems.

UAV, RF Fingerprinting, Authentication, Vibration, 5G

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1. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are increasingly pivotal in critical infrastructure, supporting applications such as military reconnaissance, environmental monitoring, medical cargo transport, and communication networks. UAVs play a central role in expanding connectivity through Space-Air-Ground Integrated Networks (SAGIN), functioning as dynamic nodes that enable low-latency communication and enhance network resilience in both remote and urban environments. With the integration of emerging 5G technologies, UAVs provide high-speed data transmission, reduced latency, and scalability, making them indispensable to the future of connected systems [1, 2].

However, the integration of UAVs into sensitive and mission-critical networks introduces significant cybersecurity vulnerabilities, emphasizing the need for robust mechanisms to

authenticate, authorize, and classify UAVs. For instance, the widely adopted MAVLink protocol has overlooked vulnerabilities in essential features like heartbeat messaging. Hughes et al. [3] propose a security design that fortifies this critical function, thereby enhancing the reliability and security of UAV communication.

Secure authorization is also fundamental to collaborative UAV movements, addressing the growing demand for high-performance communication in dynamic UAV swarms. Arya et al. [4, 5] introduce Q-Learning-based and quantum computing-driven beam reforming strategies that dynamically mitigate interference, enabling reliable collaboration and efficient swarm operations. These approaches are essential for ensuring secure and robust communication, paving the way for UAVs to meet the stringent requirements of modern interconnected systems.

As UAVs expand into critical applications, ensuring the security of their communication networks has become paramount. In addition to providing performance measurements and quality guarantees, 5G measurement parameters can serve as valuable side-channel information for enhancing UAV security. These parameters can be used for novel applications such as authentication, authorization, channel condition detection, and intrusion detection, thus improving the overall security of UAV systems. For example, Jere et al. [6] utilized Bayesian Inference-assisted Machine Learning on 5G NR data to detect and classify jamming attacks in near real-time. Similarly, Wang et al. [7] demonstrated the effectiveness of a mobile 5G testbed in IoT classification and intrusion detection, highlighting its potential for UAV security. As proposed in [8], machine learning algorithms integrated with domain knowledge can enhance the classification of LOS/NLOS signals, especially in UAV-assisted SAGIN networks. By leveraging 5G data for these purposes, UAV systems can enhance not only performance but also security through reliable identification and detection methods. Moreover, lightweight classification techniques with minimal message payload overhead are crucial for SWaP (Size, Weight, and Power)-constrained environments, ensuring energy efficiency and operational effectiveness [9]. This dual use of 5G parameters for both performance monitoring and security makes them a powerful tool for safeguarding UAV communication networks.

Among the various classification techniques, RF fingerprinting has emerged as a passive yet highly effective method for UAV identification. RF fingerprinting exploits the unique hardware-induced imperfections present in the RF signals transmitted by UAVs, allowing them to be classified without the need for active signaling or extensive data exchange. This approach has gained attention due to its minimal impact on communication overhead and the fact that it operates without altering the UAV's transmission protocols. Studies have shown the effectiveness of RF fingerprinting in ensuring the secure authentication of UAVs across a variety of environments, though its accuracy can be affected by dynamic factors

such as UAV motion [9, 10].

While RF fingerprinting holds promise, it also faces limitations, particularly when UAVs are in motion or operating in environments with high signal interference. To address these challenges, we propose investigating the use of vibration data, which offers a novel avenue for UAV classification and intrusion detection. The mechanical vibrations generated during UAV operation—caused by rotor speeds, motor dynamics, and structural characteristics—can serve as a unique identifier. These vibrations represent inherent properties of the UAV’s design and operation, potentially offering a new dimension for passive authentication and classification. This approach could be particularly advantageous in scenarios where RF fingerprinting struggles, such as high-noise environments or in the presence of significant signal overlap.

Although the use of vibration data for UAV classification is relatively unexplored, early research in structural health monitoring and performance optimization has shown the promise of vibration analysis in identifying mechanical anomalies in UAVs [11]. By extending this analysis to security applications, we aim to explore how vibration-based classification can complement existing methods like RF fingerprinting, offering a more robust solution to UAV identification in dynamic environments. Our work seeks to contribute to this emerging field by demonstrating the feasibility and benefits of using vibration data as a complementary classification method, with the potential to significantly enhance the cybersecurity of UAV communication networks.

The focus of this paper is the classification of UAVs based on their RF characteristics and mechanical vibration data, utilizing advanced deep learning models. By experimenting with both physical vibration and communication data, we aim to develop Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) that can reliably classify UAVs. One limitation of this method, as with many neural network-based approaches, is that it requires a comprehensive dataset containing samples from all known UAVs for effective training.

To ensure robustness and reliability in UAV classification, we designed our experimental settings to account for a variety of environmental conditions that could influence the collected data, particularly mechanical vibrations and 5G performance parameters. Our experiments were conducted over multiple flight sessions in different locations, including an open soccer field and an urban college campus, under varying environmental conditions such as wind speeds and levels of interference. In addition to vibration data, we utilized 5G performance measurement parameters, such as signal strength, signal-to-noise ratio (SNR), and uplink/downlink bitrates, which were collected from the UAVs equipped with 5G-enabled devices. These 5G parameters provided a crucial layer of information for RF-based classification, allowing the model to capture communication-specific features and enhance its overall accuracy.

By incorporating both mechanical vibration data and 5G communication metrics, our experimental setup captures a wide range of operational scenarios. Testing under diverse environmental conditions ensures that the machine learning models can accurately classify UAVs despite external influences like wind or structural imbalances. Moreover, the use of 5G performance data helps improve the system’s resilience by providing additional input that is less affected by environmental factors. This comprehensive approach is essential

for developing a robust UAV classification system that can generalize effectively, ensuring that the classification remains reliable across various real-world deployment scenarios.

Contributions

The key contributions of this research are succinctly summarized as follows. It is important to note that this paper utilizes experimental data from only 2 UAVs, constrained by scale and page limits. However, our methodology is not restricted to these two UAVs. Ongoing work is expanding the application of our methods to multiple UAVs under various weather conditions and other scenarios.

1. The development of a novel UAV classification system that combines both **RF characteristics** and **mechanical vibration data** to identify and authenticate UAVs, even for those of the same make and model.
2. The implementation of deep learning models, including a Recurrent Neural Network (RNN) based on vibrational signatures (achieving an accuracy of 97%) and a Convolutional Neural Network (CNN) based on 5G communication data (with an accuracy of 95%), demonstrating the feasibility of using these models for UAV classification.
3. The classification leverages measured parameters for 5G communication such as path loss, signal-to-noise ratio, and uplink/downlink bitrates, instead of conventional RF fingerprinting techniques (e.g., I/Q imbalance, phase noise) to offer a **lightweight solution** that operates efficiently in real-time. Additionally, the use of **mechanical vibration data** is a novel approach explored in this work, showing better performance compared to RF fingerprinting. This highlights its **potential for further exploration** in future UAV classification and security research.
4. The creation of a comprehensive database of UAV flight data, covering line-of-sight (LOS), non-line-of-sight (NLOS), and 5G environments, which is **openly accessible** for relevant research community.
5. The data utilized in this research, including mechanical vibration and 5G communication parameters, can be **measured independently on the UAV side**, enabling real-time, autonomous data collection and analysis for UAV classification. This paves the way for a **secure, on-chip solution** for UAV identification, enhancing the security and integrity of UAV communications in future 5G and IoT networks.

The rest of the paper is outlined as follows. Section 2 discusses the related work in the crucial role in aerial-based communication cybersecurity of UAV classification and the approaches in UAV classification. Section 3 details the experimental design of this work, followed by an explanation of the data collected and the pre-processing and analysis done to the data in Section 4. Also in Section 4 is a detailed explanation of our model configurations. Section 5 evaluates the performance of our models and discusses the data sampling used. Finally, a conclusion and consideration of future work are detailed in Section 6.

2. RELATED WORK

UAV Classification for Authentication and Authorization in Dynamic Swarm Cybersecurity

UAV classification plays a pivotal role in ensuring the security of aerial-based communication systems, particularly concerning authentication and authorization. As UAVs are increasingly integrated into both civilian and military operations, the need for robust methods to authenticate devices

and authorize access to communication networks has grown. Classification techniques are essential for distinguishing legitimate UAVs from unauthorized or rogue devices based on various UAV-specific attributes such as communication patterns, flight behaviors, and telemetry data. In particular, machine learning approaches like Support Vector Machines (SVMs) and Convolutional Neural Networks (CNNs) have proven effective in classifying UAVs. For example, Bansal and Sikdar developed the S-MAPS protocol, which leverages classification techniques to establish scalable mutual authentication for dynamic UAV swarms, ensuring that only authenticated UAVs can communicate securely within the swarm [12]. This approach addresses the security challenges posed by constantly changing swarm formations, where new UAVs may join or leave the group during operation.

Beyond authentication, UAV classification is also vital in ensuring secure authorization. In swarm-based UAV operations, different UAVs may have varying roles, such as surveillance, data collection, or command and control, and their access to network resources needs to be tightly managed. By classifying UAVs based on their roles, the system can dynamically assign different levels of access control to ensure that only authorized UAVs can perform certain tasks. In [13], the authors demonstrated an innovative radar altimeter coexistence design in the 4.2–4.4 GHz band to mitigate interference risks, ensuring the operational integrity of aviation systems amidst the expansion of 5G and beyond networks. In [14], the authors work on secure communication in fixed-wing UAV swarms demonstrates the importance of authenticated and confidential communications, where the classification of each UAV ensures that sensitive information is only accessible to trusted entities within the swarm.

The State of Art UAV Classification Approaches

Recent research in the detection and classification of UAVs explores various approaches, including radar-based, vision-based, sound-based, and Radio Frequency (RF)-based methods. Among these, RF-based methods are the most commonly adopted due to their cost efficiency and the fewer challenges they pose compared to vision and sound-based techniques [15]. RF-based classification techniques, particularly RF fingerprinting, have gained prominence because they offer a lightweight solution, with minimal computational and hardware overhead, a critical factor considering the Size, Weight, and Power (SWaP) constraints of UAVs.

RF fingerprinting works by capturing the unique hardware characteristics of UAVs, which arise from manufacturing imperfections such as variations in phase offset, clock skew, and noise in power amplifiers. These subtle differences can be learned by machine learning and deep learning algorithms to classify and authenticate individual UAVs within a network. The advantage of RF fingerprinting lies in its ability to leverage existing communication signals for classification without the need for additional sensors or significant computational resources, making it a promising approach for real-time authentication in SWaP-constrained UAV systems [16]. Moreover, deep learning techniques, particularly Convolutional Neural Networks (CNNs), have been effectively applied to RF fingerprinting tasks. CNNs have demonstrated strong performance in classifying various wireless devices by handling the spatial characteristics of RF signals [17], even though scaling these techniques to accommodate large fleets of UAVs presents challenges [18].

However, while RF fingerprinting is highly efficient in static or controlled environments, the dynamic nature of UAV flight

introduces several challenges. UAVs in motion experience Doppler shifts and channel fluctuations, which can negatively impact classification accuracy. Studies have shown that the accuracy of RF fingerprinting can decrease by up to 16% when UAVs are in motion compared to stationary conditions, particularly in environments where there are overlapping signals from other wireless sources such as WiFi and Bluetooth [10]. Furthermore, the complexity of signal interference in real-world environments necessitates the use of multi-stage classification techniques that isolate and filter UAV signals for analysis. For instance, feature selection methods like Neighborhood Component Analysis (NCA) have been employed to improve classification accuracy in these noisy conditions, achieving 93.13% accuracy when combined with a k-Nearest Neighbors (kNN) algorithm [19, 20]. In another innovative approach, RF signals have been combined with energy transient signals to classify micro-UAVs more effectively, addressing some of the challenges posed by smaller, low-power UAVs [21].

While RF fingerprinting shows great promise, its limitations in dynamic environments motivate the exploration of alternative or complementary classification features. Our work seeks to address these limitations by investigating whether mechanical vibration data generated by UAVs during flight can serve as an additional classification feature. Vibration data is influenced by factors such as motor characteristics, rotor speeds, and the structural properties of UAVs. By capturing and analyzing this data, we aim to improve classification accuracy, particularly in dynamic and noisy environments where RF fingerprinting alone may struggle. This exploration opens new avenues for multidimensional classification approaches, which can combine RF signals with vibration data to achieve more robust and reliable UAV classification, without compromising the SWaP constraints of UAV systems.

Vibration Monitoring in UAVs

Vibration monitoring in UAVs has gained significant attention, particularly in the fields of structural health monitoring and anomaly detection. UAVs are subject to various mechanical stresses during flight, such as vibrations caused by motor performance, rotor imbalances, and environmental disturbances like wind turbulence. If left unchecked, these vibrations can impact UAV stability and efficiency. Research has explored using vibration data to detect mechanical failures and maintain UAV performance in real time. For instance, in bushfire management scenarios where challenging environmental conditions exist, vibration monitoring has been critical to ensuring UAV stability and operational integrity [11]. Additionally, vibration data has been shown to affect UAV energy consumption, with excessive vibrations leading to increased power use. Monitoring these vibrations not only helps in optimizing energy use but also in prolonging the UAV's operational lifespan [22]. Moreover, there is potential to use vibration data for classification and intrusion detection, as mechanical irregularities caused by tampering or unauthorized modifications could be detected through vibration signatures. This body of research suggests that vibration analysis offers a promising path for improving UAV performance, security, and reliability in various operational contexts.

3. SYSTEM AND EXPERIMENTAL DESIGN

Our research aims to develop a machine learning model capable of accurately classifying individual UAVs using a variety of test conditions. The classification approach focuses

on RF-based data using 5G signals, not on the Pixhawk flight controller's data. We conducted flight experiments using two UAVs equipped with 5G cellular devices, enabling communication with a 5G base station. Both UAVs are of identical make and model allowing no unique physical characteristics to bias our models' decision-making allowing for better generalization across UAVs. These flights were performed under line-of-sight (LOS) and non-line-of-sight (NLOS) conditions, with and without a 5G wireless connection. Vibrational data was collected from multiple flights over several weeks at different locations, including a soccer field and an urban college campus. While environmental factors such as wind conditions varied across these tests, the use of multiple flight trials ensures that the model can learn to classify different UAVs despite the presence of these confounding factors. Data from these flights were logged by the Pixhawk flight controller and downloaded via the Mission Planner Ground Control System (GCS) to provide supplementary physical data. The overall system design is detailed in Fig. 1.

In our experiments, we also evaluated the 5G cellular network's signal strength between a ground station and the UAVs by mounting 5G-enabled mobile device at the UAV's center. It is important to note that the RF-based classification relies on 5G data collected from the cellular network, rather than the data from the Pixhawk flight controller. To maintain consistency and avoid bias from hardware variations, the same smartphone was used for all tests, focusing on analyzing the 5G signal characteristics rather than the phone's hardware fingerprint. The 5G communication data, combined with the vibrational data collected from the UAVs, was transmitted to the Ground Control System (GCS) via the base station for further analysis.

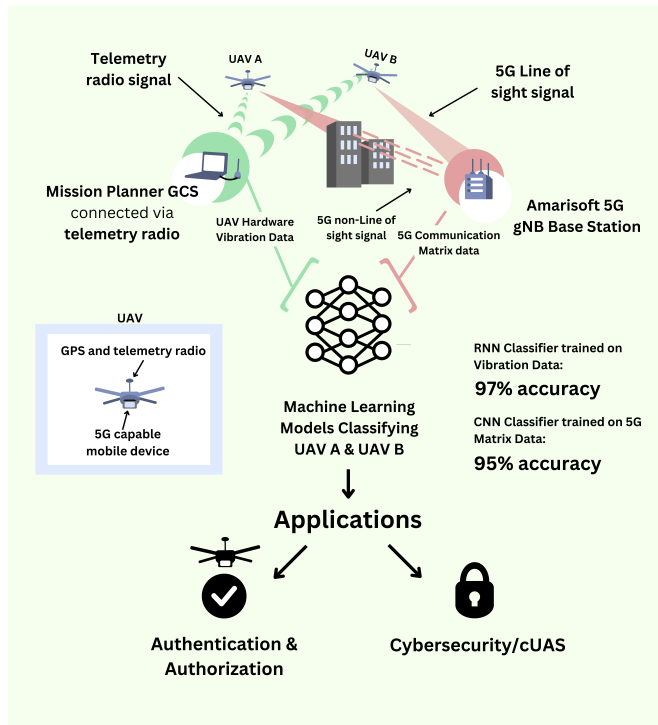


Figure 1: System Diagram - Using telemetry radio, UAV A and UAV B communicate with Mission Planner GCS. Vibration and 5G data in LOS and NLOS scenarios are collected for machine learning models with high accuracy.

Both UAV models are of identical make and model built

using the PX4 Development Kit - X500 v2, a cost-effective and lightweight UAV system that accommodates our flight controller, GPS, telemetry radios, and additional payload. The technology stack includes the advanced Holybro Pixhawk 6X FC flight controller, M10 GPS Module, a 433 MHz SiK telemetry radio for real-time data transmission to our Mission Planner GCS, and an FS-i6X 10CH 2.4GHz RC Transmitter with iA6B Receiver for flight control. Power is supplied by a 3S Lipo Battery 5200mAh 50C 11.1V, linked to a power distribution board to manage power delivery to all components.

While mechanical vibrations from the UAV are captured using onboard accelerometers, these readings may be influenced by environmental factors such as wind conditions, which can vary across flights. However, our experiments are based on multiple flight tests conducted over a range of environmental conditions, including varying wind speeds. We expect the machine learning model to generalize well and learn the classification of different UAVs, despite these confounding factors. By using diverse data from multiple flights in different environments, the model is trained to distinguish inherent UAV characteristics from those caused by external variables like wind or other environmental influences, ensuring robustness in real-world applications.

4. DATA COLLECTION AND LEARNING MODELS

The data used in our models was collected through in-flight positional and hardware telemetry, facilitated by the communication link between the Pixhawk flight controller and the Mission Planner Ground Control Station (GCS). During each flight, telemetry radios transferred data, and post-flight, the ArduPilot software on the Pixhawk stored this data in unorganized dataflash logs, lacking column headers and clear parameter descriptions.

Vibration data was captured through onboard accelerometers, which recorded responses to hardware motion, environmental factors like wind, and structural issues such as imbalances or misalignments in the UAV. This vibration data was transmitted via a 433 MHz telemetry radio and was used to train the Recurrent Neural Network (RNN). In parallel, communication data transmitted by the 5G-enabled User Equipment (UE) attached to the UAV was utilized to train a Convolutional Neural Network (CNN). By combining vibration and communication data, we aimed to create a robust classification system capable of distinguishing between UAVs under various conditions.

The pre-processing the collected data include parse and organize the disorganized dataflash logs. This approach converts the raw logs into structured files, categorizing the data into 43 distinct parameter types, such as GPS data and system status. This method significantly improved data clarity and consistency, ensuring that the data could be efficiently fed into the models while reducing manual effort and minimizing errors.

Along with the data collected from the UAV directly, we established a 5G wireless connection between a mobile device attached to the UAV and a PC running Amarisoft Next Generation Node B (gNB) base station software fielded by mounted antennas nearby on the ground. While in flight, we capture packets sent and received by our mobile device and gNB in order to analyze how the UAVs flight behavior affected the

data transfer (Fig. 2). We captured packets at varying bit rates of 1Kbps, 10Kbps, 100Kbps and 1 Mbps in multiple locations in an effort to simulate different environment scenarios. With the varying bit rates, we also strategically position the antennas to face toward and away from the general vicinity of the UAV's flight path to investigate how line-of-sight and non-line-of-sight connection affected packet retrieval.

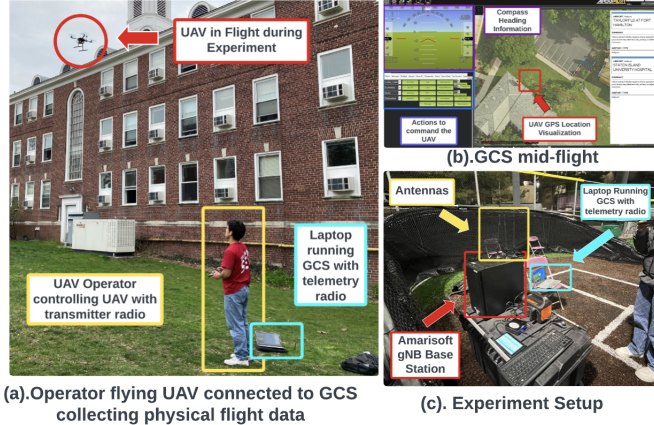


Figure 2: Experiments and Data Collection

Before selecting RNN for Vibrational Data and CNN for Communication Data, we evaluated various models including Random Forest, k-Nearest Neighbors, and Gradient Boosting. Although these models showed higher accuracy, we opted for RNN and CNN because they excel at processing time series data, which is crucial for identifying patterns in our extensive testing. Additionally, RNN and CNN are adept at capturing complex relationships within our data, which is vital for understanding the interactions between vibrational and communication data.

Vibrational Based Classification Model

In preparation for training our Recurrent Neural Network (RNN) algorithm, an essential phase of data pre-processing was undertaken to refine the dataset, focusing on including only relevant features and maintaining the data's integrity for each respective UAV instance.

In processing our dataset for optimal analysis, we concentrated on columns directly related to assessing vibrational characteristics and the data integrity of UAVs. We included three key measurements: vibration_x, vibration_y, and vibration_z. These represent the vibrational forces acting along the UAVs x, y, and z axes, providing a comprehensive view of the mechanical vibrations that could affect flight stability and component wear. Additionally, we tracked three clipping indicators—clipping_0, clipping_1, and clipping_2—which signal whether the vibrations in each corresponding axis exceed predefined thresholds, a crucial aspect in maintaining UAV operational safety and functionality. The columns 'Len' and 'crc16' were also omitted, with 'Len' indicating data length for packet transmissions and 'crc16' serving as a checksum to ensure data integrity during transmission. By focusing on these specific columns, our analysis not only monitors the UAVs vibrational state but also maintains a high standard of data reliability, crucial for developing robust predictive models. This focus on vibration data aids in predicting fluctuations in related hardware, ultimately creating a unique RF-fingerprint for each UAV, enhancing our ability to classify UAVs.

Feature	Score	Feature	Score
clipping_1	0.248523	clipping_0	0.177918
vibration_x	0.141535	clipping_2	0.139781
vibration_z	0.129838	vibration_y	0.112091
crc16	0.050313		

The annotation process involved assigning labels to differentiate the data from the two UAVs. Each data sample was annotated with a label of 0 for UAV A or 1 for UAV B. These labels provide the ground truth for the deep learning model, allowing it to learn the distinguishing characteristics between the two UAVs and predict the origin of future data samples.

To understand the contribution of each feature to the model, we used a Random Forest Classifier. This allowed us to assess feature importance by calculating the mean and standard deviation of the impurity decrease across the trees in the forest. The Len column, which contains text data, was excluded from this analysis as the classifier requires numerical inputs.

We experimented with different feature selection approaches while developing our deep learning models. Initially, the training data included three dimensions: (flights, time-steps, features). However, through testing, we found that using a simplified 2D structure, consisting of just (time-steps, features), yielded highly accurate classification. This was achieved by concatenating all flight data and using a small batch size during model training. Simplifying the feature selection process in this way reduced the amount of data required to effectively classify UAVs.

We selected the Adam optimizer due to its ability to adapt learning rates, which helped mitigate the vanishing gradient problem in our bidirectional LSTM layers. Its stability in handling gradients, combined with computational efficiency, made it well-suited for training our model on a large dataset, ensuring reliable and efficient performance.

Binary Cross-entropy was used to compile our loss function to match the nature of our data. This was chosen since we only had access to two UAVs in our experiments. In a real world application we would likely want to classify between more than two UAVs, and therefore would choose Categorical Cross-entropy instead. Other aspects of the model would also have to be tweaked should this method be used to classify between a larger array of UAVs. Currently the output layer of our model results in a number between 0 and 1. The closer the number is to 0 the more likely it is to be UAV A, and the closer to 1 the more likely it is to be UAV B. To account for additional UAVs this output layer would have to be adjusted to result in a higher dimensional vector that would represent an encoding of the likelihood of the data from a specific timestamp to belong to each of the known UAVs.

In terms of training our model, we decided to pursue common machine learning good practices like training the model via mini batch for efficient computation, using a validation set to detect overfitting and tuning hyper-parameters along with having a manageable number of epochs.

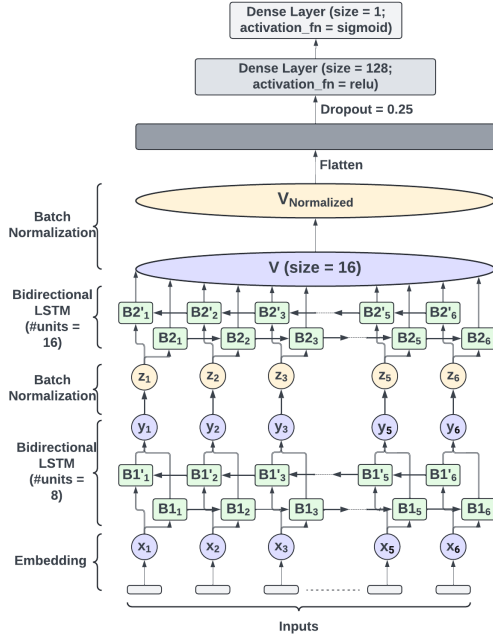
Finally, we proceeded with evaluating and classifying our model, resulting in the confusion matrix described in Table 2. The loss function utilized was binary cross-entropy:

$$-\sum_{i=1}^N [t_i \log y_i + (1 - t_i) \log(1 - y_i)]$$

where t_i is the true UAV and y_i is the predicted UAV.

Table 1: 5G Performance Measurement Parameters

Feature	Description
UL MCS	Modulation and coding scheme used for uplink transmission.
UL Path Loss	Measures signal attenuation from UAV to gNB.
Turbo Decoder Avg	Error correction performance during transmission.
DL Bitrate	Data rate from gNB to UAV (down-link).
UL PHR	Uplink packet control information for transmission management.
PUSCH SNR	Signal-to-noise ratio of the uplink shared channel.
EPRE	Energy efficiency of the transmitted signal.


Figure 3: RNN Model Summary

5G Communications Based Classification Model

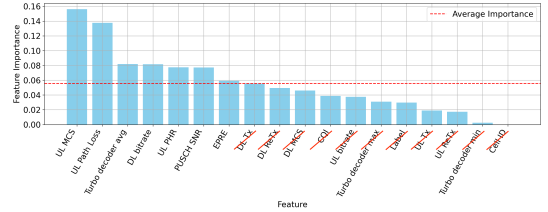
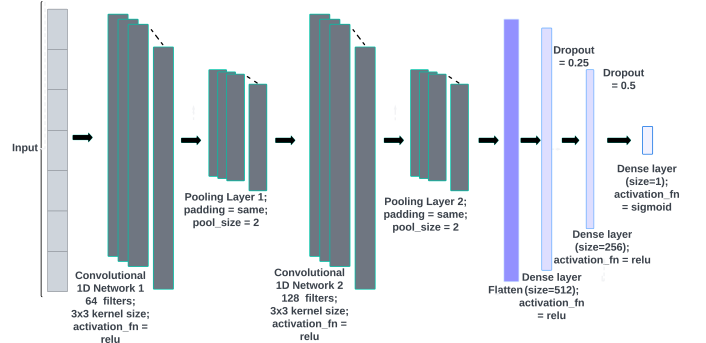
For developing our CNN we used the same method of feature selection as the physical data. The communication data has a lot more features so the Feature Importance Chart is more meaningful than it is for the vibration data. For this and the previous Feature Importance Chart we use the sklearn library for its Random Forest Classifier. The sklearn implementation allows us to easily view the feature importance by storing the calculated values in the feature_importance attribute.

The Amarisoft gNB collects various 5G communication data from the UAV's attached cell phone, capturing features related to time, bit rate, transmission, path loss, modulation, signal-to-noise ratio (SNR), and more. For our model, we selected seven features that showed importance in classification, as seen in Fig. 4: UL MCS, UL Path Loss, Turbo Decoder Avg, DL Bitrate, UL PHR, PUSCH SNR, and EPRE. These features were chosen because they are essential parameters that must be collected during communication to ensure reliable performance and quality. Moreover, these parameters introduce minimal overhead while providing sufficient information for security classification, making them efficient for real-time applications such as UAV identification.

The details of the selected 5G performance measurement parameters can be seen in Table 1.

These features provide crucial information about both the performance and signal quality of UAV communications in a 5G network. As standard communication parameters, they add minimal overhead, making them suitable for real-time security classification without compromising the efficiency of the system.

When collecting the 5G data from the UAVs, we conducted flights of both UAVs communicating with different bit-rates. We collected data from each UAV using the following bit-rates: 1 Kbps, 10 Kbps, 100 Kbps, and 1 Mbps. When developing our CNN we discovered the model performs


Figure 4: Feature Importance Chart of 5G Communication Data

Figure 5: CNN Model Summary

better when only a single bit-rate is used for the training and testing data. In theory practical applications of this technology should be able to accurately classify the same UAV when sending communications with altering bit-rates. We decided to keep the data with altering bit-rates and were still able to build a fairly accurate CNN when training with the more diverse data set.

5. PERFORMANCE EVALUATION

Data Sampling and Selection

For both our RNN and CNN models we curated a selected data set from parsing through our available data until we were left with more clearly defined sectioned t-SNE graphs. When we plotted a t-SNE graph of all our physical vibration data we found many overlapping data points from UAV A and UAV B creating an overall noisy mapping. To improve this mapping we removed all the data and one by one added data points back until we had a subsequent amount of data from training

while still maintaining clear distinctions between data from UAV A and UAV B. This dataset that plots a clear t-SNE graph we then call our ‘selected’ dataset. We followed the same process in choosing our ‘selected’ dataset for our 5G communications data. This selected data then leads both models to achieve significant improvements in their accuracy as we will discuss further below.

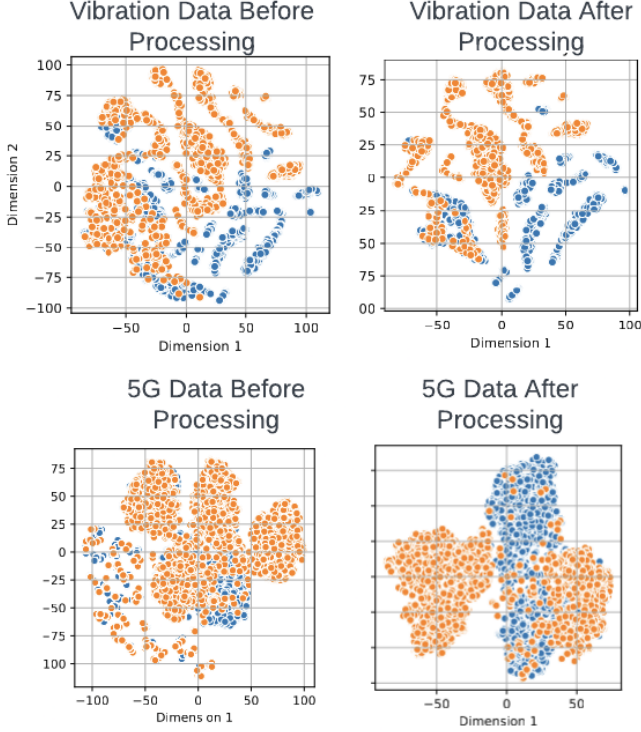


Figure 6: TSNE Comparison Before and After Data Selection

Model Comparisons

For vibration RNN Model we initially attempted to train it with all the vibration data from both UAV A and UAV B. For our final RNN model we partitioned our data on a 60% training 20% testing 10% validation split tuning our hyper-parameters to a 16 batch size over 500 epochs. Our first model boasted a 91% classification accuracy on the testing data. As stated above we then cleaned our data in an effort to gain better performance leading us to retrain our model on our selected physical vibration data. This dataset not only gave us a clearly defined t-SNE graph but also reduced training time and reaped an improved classification performance. Our final optimized model trained on this selected dataset boasts a 97% classification accuracy giving us a significantly high performing classifier.

When choosing which data logs to include in our training and testing data we used t-distributed Stochastic Neighbor Embedding (t-SNE) to visualize the high dimensional data set. This produces two dimensional vectors that can be easily plotted on a graph (Fig. 6). If we fed all the physical data we had collected into our neural network the time it took to train the network would become unreasonable. Using this t-SNE visualization we choose to include only the data that led to the clearest separation and least overlap between the data from UAV A and UAV B.

For the 5G CNN Model we also initially attempted to train

it with all our available communications between our UAVs and Amarisoft base station. For our final CNN model we partitioned our data on a 60% training, 20% testing, 10% validation split and tuned our hyper-parameters to a 32 batch size over 500 epochs. Our first model boasted a 85% classification accuracy on the testing data. As stated above we then cleaned our data in an effort to gain better performance leading us to retrain our model on our selected 5G communications data. This dataset not only gave us a clearly defined t-SNE graph but also reduced training time and reaped a significantly improved classification performance. Our final optimized model trained on this selected dataset boasts a 95% classification accuracy giving us a significantly high performing classifier.

In an effort to evaluate and compare the effectiveness of our developed models against popular models we created a visualization comparing the accuracy across epochs of our different deep learning models. Table 2 shows the comparison of the confusion matrix of the vibration data based RNN model and the 5G data based CNN model.

Table 2: Confusion Matrix Comparison

	Vibration Data RNN Model	
	predicted positive	predicted negative
actual positive	98.2%	1.8%
actual negative	5.2%	94.8%
	5G Data CNN Model	
	predicted positive	predicted negative
actual positive	95.6%	4.4%
actual negative	5.7%	94.3%

In Fig. 8 below, we show the accuracy of our RNN when trained on all our data collected, as well as when only trained on the selected set of data, as described in Section 6.2. We also graph the CNN when trained on all data as well as when only trained on the data where all communication features are transmitted with the same bit-rate. In addition we also include graphs of the accuracy across several classical machine learning classifiers fitted to the vibration data.

Fig. 7 also shows the accuracy of the validation data rather than training data. This is done since the validation data is a subset of the training data and is analogous to the accuracy of the model on the testing data.

As you can see from Fig. 7 and Fig. 8 the RNN, trained on the selected dataset, performs the best. Followed by the CNN, trained on the same bit-rate, then the RNN trained on all the data. The multimodal network only performs better than the CNN, trained on different bit-rates. This indicates that these models need more adjustments before they can be used for a practical application. It also shows that these neural networks are learning from the intrinsic characteristics of the UAV hardware and the 5G signal to make an accurate classification. Further refinements to these models could allow them to perform as well as the classic machine learning classifiers.

6. CONCLUSION AND FUTURE WORK

In this work, we proposed a novel approach to UAV classification by combining RF characteristics and mechanical vibration data, allowing for accurate identification of UAVs of the same make and model. Through extensive Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) tests, we developed a dataset incorporating both communication and

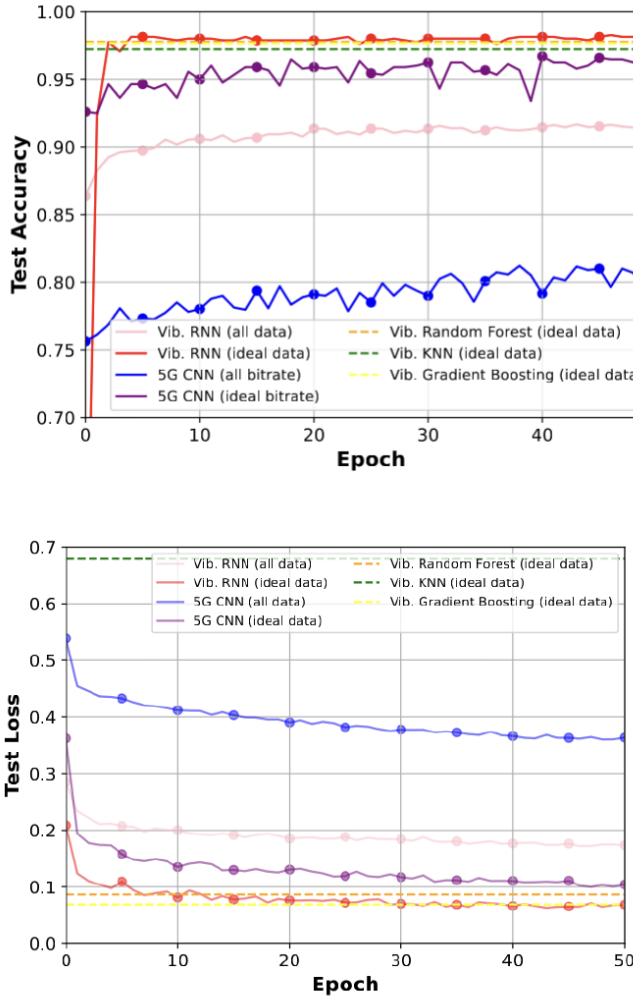


Figure 7: Test Accuracy and Test Loss Performance vs the Epoch

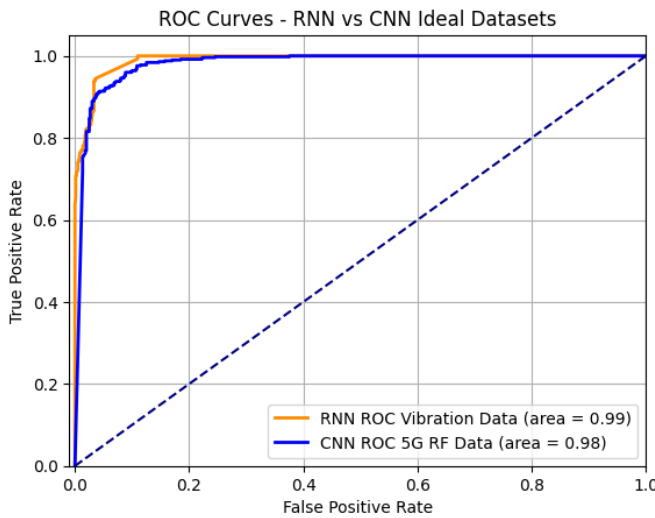


Figure 8: ROC curves for CNN model and RNN model using ideal dataset

physical data, which was used to train deep learning models. Our models demonstrated high accuracy in distinguishing

between UAVs of the same type, showcasing the potential of integrating mechanical vibrations with RF signals to enhance UAV classification capabilities. This approach addresses the limitations of conventional RF fingerprinting methods, particularly in dynamic environments. The combination of deep learning with in-flight data provides a promising direction for enhancing UAV security and authentication.

Moving forward, our research will involve collaboration with the Aerial Experimentation and Research Platform for Advanced Wireless (AERPAW), a wireless research platform led by North Carolina State University. This will enable us to extend our experiments on a larger scale by accessing telemetry data from various UAV architectures under real-world conditions. This platform will allow us to further validate our models across different UAV designs and environments, improving the robustness of our classification system. Additionally, future research will explore refining the deep learning models for real-time implementation, with a focus on minimizing computational overhead and integrating this classification system into a secure, on-chip solution for UAV communications.

7. ACKNOWLEDGEMENT

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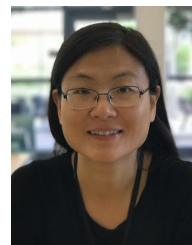
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